CISPA
HELMHOLTZ CENTER FOR INFORMATION SECURITY

Cryptography

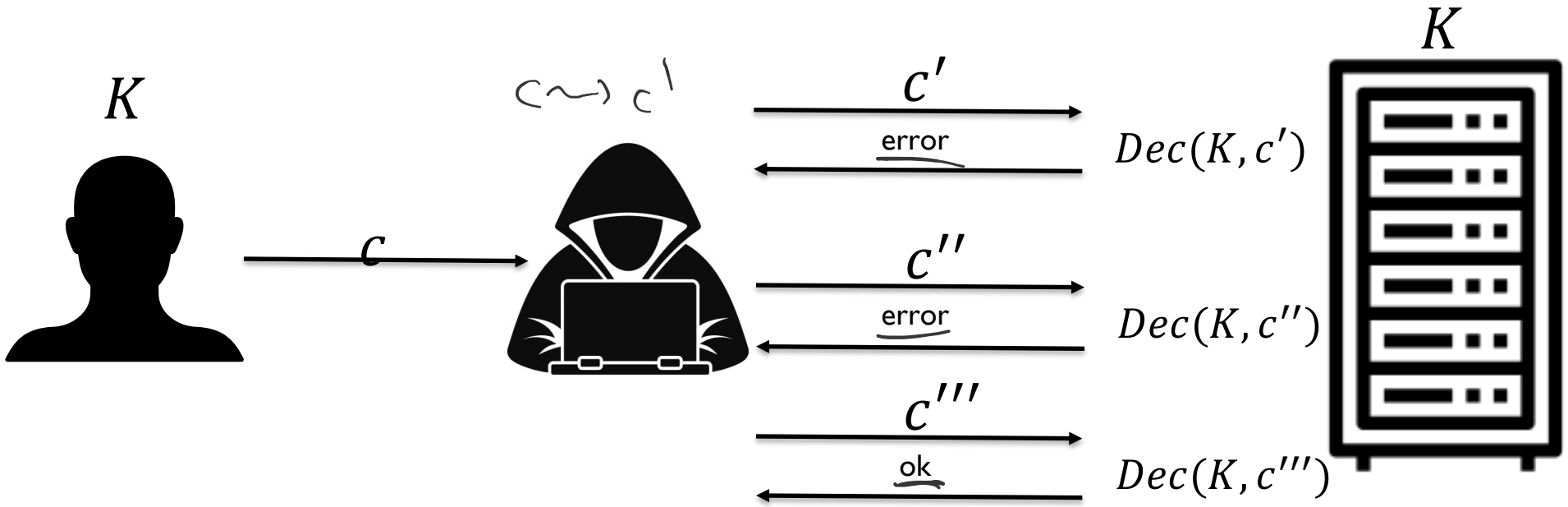
Lecture 10: Chosen Ciphertext Security

Nico Döttling

Encryption and Authentication

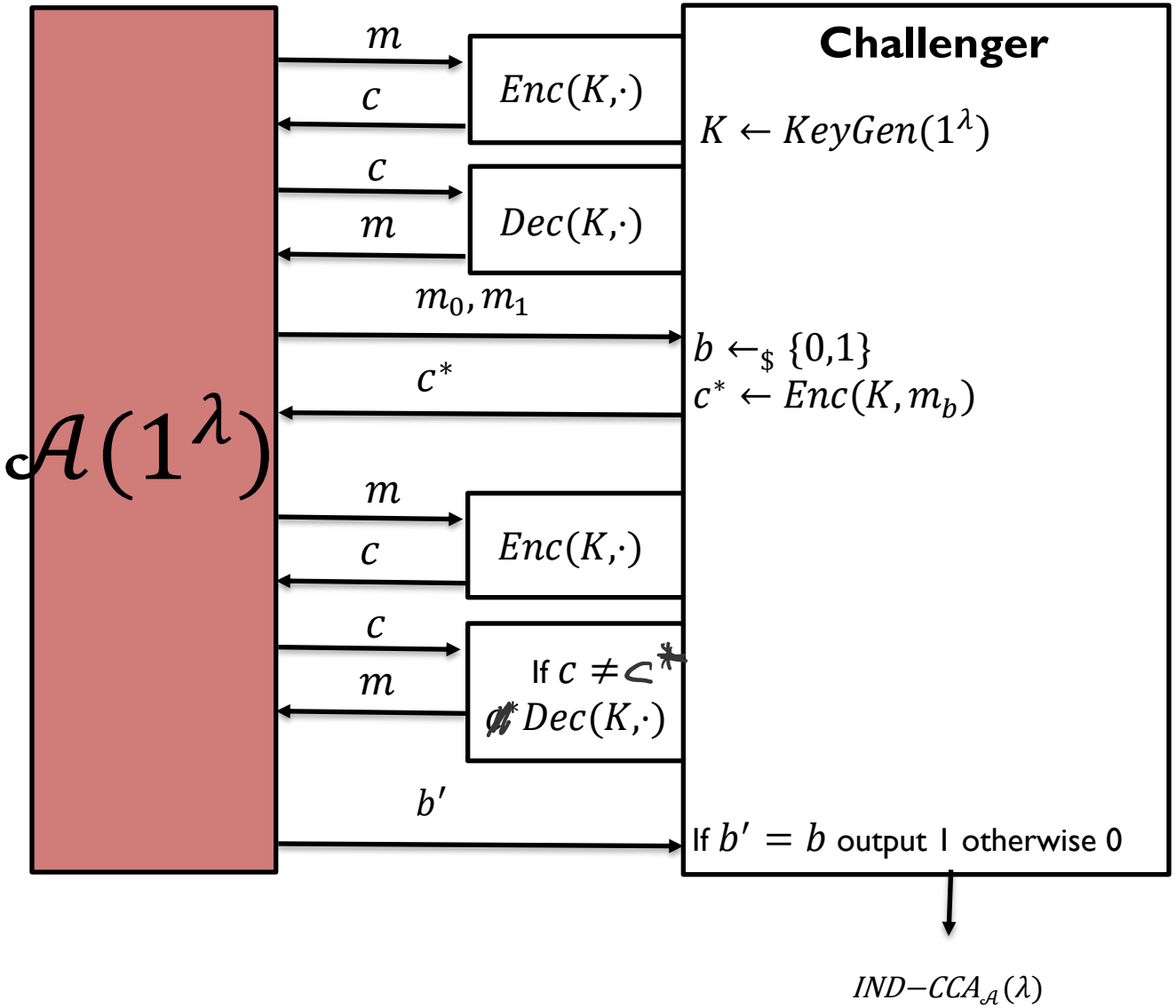
- …are different goals…
- …not always!

Man-in-the-Middle Attack
against secrecy



Chosen Ciphertext Security

$IND{-}CCA_{\mathcal{A}}(\lambda)$ Experiment:



Definition

An encryption scheme $(KeyGen, Enc, Dec)$ is indistinguishable under chosen ciphertext attacks, or $IND{-}CCA$-secure, if it holds for every PPT-bounded adversary $\mathcal{A}$ there exists a negligible function ν s.t. for all $\lambda \in \mathbb{N}$

$$Pr[IND{-}CCA_{\mathcal{A}}(\lambda) = 1] < 1/2 + \nu(\lambda)$$

Constructing IND-CCA secure encryption

- Idea: Use authentication
- But how?
- 3 Options:
  1. Authenticate and Encrypt: $Enc(K_1, m), Mac(K_2, m)$
  2. Authenticate then Encrypt: $Enc(K_1, m \parallel Mac(K_2, m))$
  3. Encrypt then Authenticate: $\underbrace{Enc(K_1, m)}_{c}, Mac(K_2, \cancel{Enc(K_1, m)})$ c

Constructing IND-CCA secure encryption

1. Authenticate and Encrypt: $Enc(K_E, m), Mac(K_M, m)$
   - Is bad: $Mac(K_M, m) = (Mac'(K_M, m), m)$
2. Authenticate then Encrypt: $Enc(K_E, m \parallel Mac(K_M, m))$
   - Also bad: $Enc(K_E, m) = (Enc'(K_E, m), r)$
3. Encrypt then Authenticate: $Enc(K_E, m), Mac(K_M, Enc(K_E, m))$
   - Winner!

Constructing IND-CCA secure encryption

Construction: Let $(KeyGen, Enc, Dec)$ be an encryption scheme and $(Gen, Mac, Verify)$ a MAC

- $KeyGen'(1^\lambda)$: Compute $K_E \leftarrow KeyGen(1^\lambda)$ and $K_M \leftarrow Gen(1^\lambda)$ and output $K \leftarrow (K_E, K_M)$
- $Enc'(K, m)$: Parse $K = (K_E, K_M)$. Compute $c \leftarrow Enc(K_E, m)$ and $t \leftarrow Mac(K_M, c)$. Output $c' \leftarrow (c, t)$
- $Dec'(K, c')$: Parse $K = (K_E, K_M)$ and $c' = (c, t)$. If $Verify(K_M, c, t) = 1$ output $m \leftarrow Dec(K_E, c)$, otherwise $\perp$

Correctness: Follows from correctness of $(KeyGen, Enc, Dec)$ and $(Gen, Mac, Verify)$

<u>Theorem 6.5</u>

If $(KeyGen, Enc, Dec)$ is an IND-CPA secure encryption scheme and $MAC = (Gen, Mac, Verify)$ a sEUF-CMA secure message authentication code, then $(KeyGen', Enc', Dec')$ is an IND-CCA secure encryption scheme

Proof Idea

Summary

- Active adversaries try to break secrecy by manipulating messages and observing behavior other parties
- IND-CCA security provides security against active adversaries
- Modeled by giving the adversary access to a decryption oracle
- IND-CCA encryption from IND-CPA encryption and MAC via encrypt-then-mac!